

Executive Business Report

13th July, 2026

Sales Forecasting & Demand Intelligence System

Prepared for: Head of Supply Chain & Chief Financial Officer (CFO)

1. Executive Summary

This report presents the key findings from the Sales Forecasting and Demand Intelligence System developed using historical retail sales data. The system analyzes sales trends, predicts future demand, detects unusual sales patterns, and groups products based on demand levels. Among the forecasting models tested, **XGBoost** delivered the highest accuracy and was selected as the best-performing model. These insights can help improve inventory planning, reduce stock shortages, and support better business decisions.

2. Key Findings from Data Analysis and Forecasting

- Sales showed clear seasonal trends, with higher demand during specific periods.
- Technology products generated the highest revenue, while Office Supplies maintained consistent sales.
- Regional sales performance varied, highlighting the need for region-specific inventory planning.
- Among the forecasting models, **XGBoost** achieved the best performance with the lowest forecasting error (MAPE: **14.79%**).

3. Three-Month Sales Forecast

Three forecasting models were evaluated:

Month	Forecasted Sales
Month 1	85,866
Month 2	86,568
Month 3	88,749

The forecast indicates a stable increase in sales over the next three months. Actual sales are expected to remain within approximately **±15%** of these predictions under normal business conditions.

4. Top Anomalies Detected

- **High Sales Spike:** Likely caused by seasonal demand or promotional campaigns.
- **Low Sales Period:** Possibly due to inventory shortages or reduced customer demand.
- **Unusual Sales Pattern:** May indicate bulk purchases, pricing changes, or unexpected market behavior.

These anomalies should be monitored to improve forecasting accuracy and business planning.

5. Product Demand Segmentation

- **High-Demand Products:** Maintain higher inventory and prioritize replenishment.
 - **Medium-Demand Products:** Keep a balanced stock and monitor sales regularly.
 - **Low-Demand Products:** Reduce inventory levels and use promotional offers to improve sales.
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6. Business Recommendations

1. **Use XGBoost for demand forecasting** since it achieved the highest prediction accuracy.
 2. **Adopt demand-based inventory planning** by stocking products according to their demand segment.
 3. **Monitor sales anomalies regularly** to quickly identify unusual business events and adjust inventory strategies.
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7. Risk / Limitation

The forecasting system relies on historical sales data and may not fully capture unexpected events such as economic changes, major promotions, or supply chain disruptions. Regular model updates with new data will improve its reliability.

8. Conclusion

The Sales Forecasting and Demand Intelligence System provides valuable insights for sales prediction, anomaly detection, and inventory planning. By implementing the recommended strategies, the company can improve demand forecasting, optimize stock levels, reduce operational costs, and make more informed business decisions. The system serves as a practical decision-support tool for supply chain and financial planning.

Demo Link: <https://salesforecasting-xylofy-ai.streamlit.app/>

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